

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads	)	Docket No. RM26-4-000
to the Interstate Transmission System	)	

**REPLY COMMENTS OF THE PEW CHARITABLE TRUSTS, ENERGY
MODERNIZATION PROJECT ON THE ADVANCE NOTICE OF PROPOSED
RULEMAKING**

The Pew Charitable Trusts (Pew) submits these reply comments in response to the Advance Notice of Proposed Rulemaking (ANOPR) on Ensuring the Timely and Orderly Interconnection of Large Loads.¹ Pew’s Energy Modernization Project works with state and federal policymakers to build a modern, reliable, and affordable electric grid, including the adoption of local, distributed energy, such as home solar and batteries.

While commenters disagree on the question whether the Commission should take jurisdiction over the interconnection of large load customers to the transmission system (a question which Pew takes no position on), they are united on the imperative to speed up interconnection processes while preserving affordability for existing customers.² Many commenters endorse the U.S. Department of Energy’s (DOE’s) proposed principles designed to achieve these goals: coordinated study of load and generation, as well as expedited interconnection for large load customers and hybrid facilities.³

In addition, the evidence in the record demonstrates that Virtual Power Plants (VPPs) can be integrated into implementation of these principles, and that providing for

¹ *Ensuring the Timely and Orderly Interconnection of Large Loads*, Advance Notice of Proposed Rulemaking, Docket No. RM26-4-000 (Oct. 23, 2025) (ANOPR).

² *See, e.g.*, Amazon Energy Comments at 3; Duke Energy Comments at 6; EEI Comments at 2; EsVolta Comments at 5; Exelon Comments at 3; Maryland Public Service Commission Comments at 5, 16; Microsoft Comments at 9, 11; Shell Energy North America Comments at 11.

³ *See, e.g.*, DTE Electric Comments at 10-11; EDF Power Solutions Comments at 12, 16; ENGIE Comments at 5, 9; Google Comments at 6-7; Meta Comments at 4-5; Shell Energy North America Comments at 4, 6-7.

large load customers and hybrid facilities to be able to contract with VPPs and to have those contracts accounted for within the scope of interconnection procedures would greatly further DOE’s goals, while advancing the Commission’s statutory mandate and principles of facilitating resource-neutral competition in wholesale markets. Including VPPs within the scope of the rule is consistent with Commission precedent dictating that aggregations of Distributed Energy Resources (DERs) should be permitted to offer services that they are technically capable of providing, and could be done without expanding on the scope of the Commission’s jurisdictional claim or intruding upon state-jurisdictional VPP programs.⁴ Given the scale of the challenge that DOE and the Commission face in facilitating speed-to-power for large load customers while keeping the system affordable for existing customers, the Commission cannot afford not to deploy this important tool. Rather, it should insist that transmission providers include it within the toolbox and use it for the job at hand.

I. TO THE EXTENT THE COMMISSION ASSERTS JURISDICTION IN THIS PROCEEDING, IT SHOULD ADOPT RULES TO ENABLE INTEGRATED, EXPEDITED INTERCONNECTION STUDY OF LARGE LOADS CONTRACTING WITH VIRTUAL POWER PLANT OPERATORS TO CURTAIL OR DISPATCH.

A. Accounting for contracts between large load customers and VPPs in the interconnection process could help to expedite interconnection for large load customers while reducing system costs for all customers, goals that are widely supported by commenters.

The ANOPR proposes that “[l]oad and hybrid facilities should be studied together with generation facilities,”⁵ and states that “FERC should expedite interconnection study of large loads that agree to be curtailable and hybrid facilities that agree to be curtailable and

⁴ *Participation of Distributed Energy Res. Aggregations in Mkts. Operated by Reg’l Transmission Orgs. & Indep. Sys. Operators*, Order No. 2222, 172 FERC ¶ 61,247, at P 26 (2020), *order on reh’g*, Order No. 2222-A, 174 FERC ¶ 61,197, *order on reh’g*, Order No. 2222-B, 175 FERC ¶ 61,227 (2021).

⁵ ANOPR at P 20.

dispatchable.”⁶ To the extent that it asserts jurisdiction over large load and hybrid facility interconnection through this proceeding, the Commission should adopt and clarify these proposals. In doing so, the Commission should ensure that large load customers are able to contract with VPPs that curtail or dispatch. A broad range of commenters support including VPPs within the scope of the Commission’s rules,⁷ emphasizing how VPP’s can be a critical tool in achieving DOE’s objectives and supporting the Commission’s statutory mission of ensuring just and reasonable and not unduly discriminatory rates. Adopting rules that allow integrated, expedited study of large loads that contract with VPPs will facilitate speed-to-power for large load customers while keeping transmission and energy affordable for existing customers.

Secretary of Energy Wright’s letter directing the Commission to consider the ANOPR emphasizes the need to “ensure efficient, timely and non-discriminatory load interconnections,” while guaranteeing that “all Americans and domestic industries have access to affordable, reliable, and secure electricity.”⁸ Achieving these twin goals simultaneously is challenging because serving large loads requires the rapid addition of new capacity resources and transmission system upgrades, which are costly and time consuming to build. Under the status quo, *none* of the Secretary’s objectives will be realized: large loads will face continuing delays connecting to the grid while electricity becomes more expensive and less reliable for existing

⁶ *Id.* P 24.

⁷ See Equinix, Inc. Comments at 7 (“[W]e suggest that large load interconnection customers should be allowed to include what could be termed ‘aggregated resource’ proposals in their interconnection requests”); Rewiring America Comments at 2-3; Distributed Capacity Parties Joint Comments at 6 (“Operationally, a hyperscaler could contract with a battery installer to develop new batteries in the area of a relevant transmission and/or capacity constraint.”); Solar Energy Industries Association Comments at 14 (recommending that in implementing principle 3, “[t]he Commission should . . . make explicit that BYOG arrangements may include Distributed Energy Resource Aggregations (sometimes referred to as ‘Virtual Power Plants’ (VPPs).”); Tesla Comments at 4 (“Tesla urges the recognition of DERs and VPPs as meaningful grid service tools that can support faster interconnection of large loads to the transmission system. VPPs should be considered for their potential to alleviate transmission network constraints and capacity needs associated with the interconnection of large loads.”).

⁸ Secretary of Energy Chris Wright, *Secretary of Energy’s Direction that the Federal Energy Regulatory Commission Initiate Rulemaking Procedures and Proposal Regarding the Interconnection of Large Loads Pursuant to the Secretary’s Authority Under Section 403 of the Department of Energy Organization Act*, at 1-2 (filed Oct. 27, 2025).

customers. For example, comments filed from Distributed Capacity Parties emphasize that “[l]arge loads face interconnection delays for several reasons, including insufficient capacity to supply new load reliably,” and “the need for costly network upgrades to bring incremental peak power to the location of new load.”⁹

A large majority of commenters who address large load flexibility support the concept of expedited interconnection for curtailable large load customers and curtailable and dispatchable hybrid facilities. AEMA tallies more than a hundred comments in support as compared to just over a dozen “cautionary voices.”¹⁰ Enabling flexibility, in practice, however, will require an all-hands-on-deck approach.¹¹ This includes utility-scale hybrid generation facilities and energy storage,¹² demand response,¹³ VPPs and DERs,¹⁴ and data center flexibility management software solutions.¹⁵ The exact mix depends on the type of large load facility and its unique circumstances.

Given the scale of large load growth, and the variety of contexts in which large loads may seek interconnection service, it is important to provide as many tools as possible to enable flexibility—thereby unlocking more megawatts (MWs) of flexibility and making flexibility available to a broader set of commercial entities. The scale of the opportunity that VPPs could provide to support large load flexibility is significant: “In 2023 alone,” for example, Distributed Capacity Parties note that “Southwest Power Pool (‘SPP’) added approximately 431 MW new

⁹ Distributed Capacity Parties Joint Comments at 4.

¹⁰ See AEMA Reply Comments at 1.

¹¹ See Equinix Comments at 7 (urging Commission to enable “a wide range of flexible resources” given that not all data centers may be located in areas that are amenable to onsite generation and/or storage, and some may have strict minimum security risk and data latency requirements).

¹² See, e.g., Clean Energy Buyers Association Comments at 15; EsVolta Comments at 4; L. Lynne Kiesling, Ph.D. Comments at 6; U.S. Energy Storage Coalition Comments at 1-2.

¹³ See, e.g., Load Flexibility Parties Joint Comments at 2-4; Solar Energy Industries Association Comments at 4.

¹⁴ See, e.g., Rewiring America Comments at 1; Tesla Comments at 4.

¹⁵ See, e.g., Emerald AI Comments at 1; L. Lynne Kiesling, Ph.D. Comments at 6.

demand response resource capability, equivalent to a large power plant.”¹⁶ Similarly, Rewiring America emphasizes that “[h]yperscalers could meet one third of their projected additional capacity needs by paying for heat pumps in select households that currently use inefficient electric heating, cooling, and water heating,” and “could more than meet their total planned capacity needs by paying for battery storage as well as rooftop solar for homes well suited for it.”¹⁷ In addition, VPPs can be integrated with other solutions. As Emerald AI explains in their comments, VPPs “complement . . . -[other] types of data center flexibility to . . . augment the total flexibility a facility can offer.”¹⁸ This is especially true for data centers “located in very close proximity to financial sector, national security, or other critical uses that demand the absolute minimum security risk and data latency.”¹⁹ By allowing “[t]hose data centers located in dense metropolitan areas . . . to tap into the wealth of flexibility opportunities that surround them,”²⁰ the Commission can facilitate speed-to-power and consumer protections that might not be available if the Commission limited hybrid facility fast tracks only to large loads contracting with utility-scale resources.

In addition to enabling simpler and faster interconnection study, VPPs can facilitate speed-to-power because they are much faster to deploy than utility-scale infrastructure, minimizing potential construction and supply chain delays that have exacerbated resource adequacy challenges and transmission system headroom scarcity. As commenters explain, the core challenge facing transmission providers across the country is that new load is being added to the system faster than new supply can interconnect. While grid operators across the country

¹⁶ Distributed Capacity Parties Joint Comments at 3.

¹⁷ Cora Wyent et al., *Homegrown Energy: How household upgrades can meet 100 percent of data center demand growth*, REWIRING AMERICA, at 5 (2025), https://a-us.storyblok.com/f/1021068/x/1346f9b42e/homegrown-energy-report_-rewiring-america.pdf?cv=1761923965791.

¹⁸ Emerald AI Comments at 4.

¹⁹ Equinix, Inc. Comments at 7.

²⁰ *Id.*

are focused on accelerating the pace of their interconnection processes, queues remain clogged and appear likely to continue to remain clogged at least in the near term. VPPs, by contrast, do not depend on proceeding through the large generator interconnection queue. As Distributed Capacity Parties explain, “[d]istributed capacity assets are installed today across critical footprints, are readily installable en masse, and may be aggregated into utility-scale assets on a *timeline of months rather than years*” (emphasis added).²¹

B. Commission precedent supports including VPPs within the scope of any interconnection rule, because they are technically capable of providing the same system value as direct curtailment or dispatch of the large load or hybrid facility.

VPPs are technically capable of providing the same interconnection service benefits as direct facility curtailment and dispatch. As such, Commission precedent supports including them within the scope of any rule providing for joint or coordinated study of load and generation, and/or expedited interconnection for curtailable and dispatchable facilities.

As Rewiring America emphasizes, “[o]ff-site distributed energy resources (‘DERs’) can deliver the same types of energy services—generation, load reduction, or transmission relief—that on-site assets provide.”²² From the resource adequacy perspective, Rewiring America notes that “demand-side measures that permanently lower peak load[,]” which can include both reliance on DERs to enable curtailed consumption from network resources and coordinated distributed demand reduction to offset large load needs, “can meaningfully reduce the system impacts of new large loads.”²³ Distributed Capacity Parties also note the ability of virtual power plants to increase transmission system utilization and save on network upgrades, including through the use of “‘latent’ demand flexibility [to] meet incremental capacity needs by reducing

²¹ Distributed Capacity Parties Joint Comments at 3 (emphasis added).

²² Rewiring America Comments at 2.

²³ *Id.* at 3.

power draw from the grid, thereby freeing up network capacity.”²⁴ In order to provide these transmission system benefits, Rewiring America highlights that “[l]ocational requirements can ensure these resources provide benefits on the same part of the grid affected by the interconnection.”²⁵ The key point is that when VPPs curtail and dispatch, this can provide the same system benefits as when a large load does so directly. As Rewiring America points out, “[W]hat matters is equivalent, verifiable performance in mitigating constraints within the same constrained transmission pocket or local grid area, not whether the asset is physically inside or outside the facility fence line.”²⁶ In a similar vein, Equinix explains, “[s]o long as a large load is contracted with and can deliver a solution that resolves or mitigates the constraints and violations that would otherwise trigger incremental energy system costs, that aggregated proposal should be studied as part of the large load interconnection request.”²⁷

The same principles underlying Order No. 2222 require the Commission to provide the opportunity for VPPs to participate in hybrid and flexible arrangements where technically capable of doing so. Specifically, to the extent that the Commission adopts rules for the combined or coordinated study of generation and loads, it should require transmission providers to facilitate a pathway through which VPPs can be accounted for in load interconnection studies on a comparable basis to utility-scale generation.

In Order No. 2222, the Commission found that existing regional transmission organization (RTO)/independent system operator (ISO) market rules were unjust and unreasonable because they “present[ed] barriers that prevent[ed] certain distributed energy resources that are technically capable of participating in the RTO/ISO markets on their own or

²⁴ Distributed Capacity Parties Joint Comments at 6.

²⁵ Rewiring America Comments at 3.

²⁶ *Id.*

²⁷ Equinix, Inc. Comments at 7.

through aggregation from doing so[.]” which “reduce[d] competition and fail[ed] to ensure just and reasonable rates.”²⁸ The same non-discrimination principle requires the Commission to allow aggregated DERs that are technically capable of providing hybrid service to avail themselves of the opportunities available to utility-scale hybrid facilities. As Equinix explains, “[f]rom a grid operations, transmission upgrade, or incremental energy need perspective, it is irrelevant whether a large load is curtailed, backup or other co-located generation reduces the customer’s draw from the grid, or other load, generation or storage in sufficient proximity reduces draws from the grid.”²⁹ Just as opening wholesale markets to DER aggregations “enhance[s] the competitiveness, and in turn the efficiency, of RTO/ISO markets and thereby help[s] to ensure just and reasonable and not unduly discriminatory or preferential rates for wholesale electric services[.]”³⁰ “[e]nabling a wide range of flexible resources” including VPPs “to be paired with large load interconnection requests would . . . spark competition and innovation” and “enable lower overall costs and potentially broader benefits.”³¹ Accordingly, the Commission should extend the logic of Order No. 2222 to any hybrid-specific large load interconnection rules that it adopts and allow all resources technically capable of providing the relevant service to participate.

Neglecting to do so and thereby excluding resources that are similarly situated in their technical ability to provide the relevant service and benefits from participating in the hybrid-specific procedures would clearly raise questions of undue discrimination.

C. Rules leveraging the value of technically-capable VPPs to facilitate interconnection are compatible with a broad range of implementation frameworks set forth by commenters.

²⁸ Order No. 2222, 172 FERC ¶ 61,247 at P 26.

²⁹ Equinix, Inc. Comments at 6-7.

³⁰ Order No. 2222, 172 FERC ¶ 61,247 at P 29.

³¹ Equinix, Inc. Comments at 7.

Accordingly, if the Commission adopts rules providing for joint or coordinated study of large loads with generation supply and expediting integrated interconnection study for large load and hybrid resources it should require transmission providers to enable VPPs to contribute and be accounted for in these rules. **Specifically, the Commission should *define hybrid facility in a manner that includes large loads paired with VPPs and/or allow for large load customers and hybrid facilities to contract with VPPs or coordinate operations with jointly-owned VPPs in order to provide curtailment and/or dispatch capability.*** This point stands regardless of how the Commission proceeds from the ANOPR, whether fully standardizing large load interconnection procedures, adopting a more limited set of requirements specific to flexible loads, or adopting even more limited requirements applicable only to hybrids.

Commenters have pointed out that two types of impacts of large load and hybrid facility additions may be accounted for through this rule: (1) the impact of these facilities on supply resource needs; and (2) the impact of these facilities on the transmission system. To the extent that the Commission addresses either type of impact, VPPs can and should be leveraged to offset both of these types of impacts in a large load interconnection rule.

VPPs' greatest value in this context may be their ability to offset the supply resource needs created by large load customers (e.g. Network Resource requirements in order to interconnect and obtain transmission service, as well as resource adequacy requirements). Various proposals recommend jointly studying or providing for coordinated study of large load customers with associated capacity supply, expediting interconnection of resources that bring their own capacity, or conditioning interconnection of large loads on bringing their own capacity.³² Should the Commission adopt such a proposal or develop a similar framework, it

³² Land Trust Alliance at 10 (recommending that load interconnection be conditioned upon curtailment commitments or addition of new generation by load to meet their energy needs); Pennsylvania Office of Consumer Advocate and

should ensure that VPPs are eligible to provide capacity for these interconnection study purposes.

For example, Industrial Customer Organizations advocate for the implementation of an elective “Fast Lane Interconnection” for customers who, among other criteria, “can add new generation to the system”³³ Similarly, Critical Loop recommends that the Commission “[r]ecognize [Large Load Flexibility Commitments (LLFCs)] for capacity/resource adequacy (capacity accreditation or reduced procurement obligations), enabling faster energization when capacity constraints would otherwise delay interconnection.”³⁴ Distributed Capacity Parties suggest that “[n]ew large loads that bring their own capacity—in whatever form—should be studied together with that capacity.”³⁵ As Critical Loop and Distributed Capacity Parties emphasize, these proposals could be implemented to “explicitly include study of *incremental resources in the relevant grid area*, whether or not those resources look like traditional utility-scale power plants.”³⁶ Aggregated DERs across the relevant balancing authority are capable of reducing a large load customer’s capacity impacts in any such cases and should be provided for in any resulting rule.

To the extent that the Commission takes action in this proceeding, its approach should recognize that capacity supply constraints prompted by large load additions to the system are a key part of the need for reform.³⁷ Likewise, the Commission should recognize the multiple

Delaware Division of the Public Advocate Comments at 24 (“The Joint Consumer Advocates support large loads and hybrid facilities that interconnect to the system only if they bring sufficient capacity (either through generation and reserves or capacity-backed demand response) to offset the additional load brought onto the grid.”); Virginia State Corporation Commission at 3 (“The Commission should enable the accelerated interconnection of large loads that are willing to bring their own generation”).

³³ Industrial Customer Organizations Joint Comments at 5.

³⁴ Critical Loop Comments at 3.

³⁵ Distributed Capacity Parties Joint Comments at 5.

³⁶ *Id.* at 5-6 (emphasis in original).

³⁷ See Independent Market Monitor for PJM, Complaint, Docket No. EL26-30-000, at 3-4 (filed Nov. 25, 2025) (“Data center load growth is the primary reason for recent and expected capacity market conditions, including total

benefits that VPPs can offer. While some commenters argue that expedited and integrated load interconnection processes for flexible loads contracting with new supply resources should be divorced from consideration of those large load and hybrid resources' resource adequacy impacts,³⁸ these perspectives overlook the unique system challenges that the Commission now faces. In today's market conditions, there is a critical link between generator interconnection study speed and resource adequacy challenges.

Interconnection rules related to resource adequacy impacts and ensuring that balancing authorities have adopted operations protocols to enable these interconnection frameworks are distinct from adopting any nationally-applicable rules for the resource adequacy frameworks themselves. And while it may be challenging for the Commission to include express requirements for any modifications to resource adequacy frameworks as part of this ANOPR given the widely varying nature of resource adequacy requirements and the state-jurisdictional nature of these requirements in some regions, the Commission could nevertheless encourage regions with Commission-jurisdictional capacity or resource adequacy frameworks to consider related reforms as part of compliance.

forecast load growth, the tight supply and demand balance, and high prices . . . existing and forecast data center load by itself resulted in a significant increase in the 2026/2027 BRA revenues . . . This total will continue to grow until the issues associated with the additions of large data center loads are addressed.”); MISO, Filing, Docket No. ER25-2454-000, at 5 (filed June 6, 2025) (noting that “[u]rgent resource adequacy and reliability challenges are arising because of . . . unexpected demand for energy-hungry data centers to support artificial intelligence” among other causes); SPP, Filing, Docket No. ER25-2296-000, at 1-2 (filed May 22, 2025) (explaining that “The SPP region is on the precipice of a resource adequacy crisis” in part because “A sudden increase in large loads from data centers and other technologies have further accelerated already increasing demand.”); *see also* Denise Foster Cronin on Behalf of East Kentucky Power Cooperative Prefiled Statement, Docket No. AD25-7-000, at 2 (filed May 16, 2025) (“Our concern is that as the PJM region’s supply and demand balance tightens and potentially falls short of the Reliability Requirement, entities like EKPC who endeavor to match their portfolio of owned and bilaterally contracted resources to their forecasted load needs will face additional, unreasonable load shed risk . . . This concern becomes particularly acute as additional large loads, like data centers or large manufacturing, come on line without the resources necessary to serve them being secured.”); Eolian, L.P Comments, Docket No. AD25-7-000, at 1 (filed July 7, 2025) (“Data centers have created significant load growth for the first time in decades and although Transmission Providers have made positive queue-reform efforts to increase the speed by which studies are completed and a GIA is achieved, those . . . reforms do not provide means for emerging load to be matched with new supply resources including generation and storage.”).

³⁸ *See* Constellation Energy Corporation Comments at 15-17; Vistra Comments at 29-31.

Existing resource adequacy constructs may no longer be producing just and reasonable rates because generator interconnection queue delays prevent new supply from responding to high prices,³⁹ and integrated reform of both sets of rules may be appropriate. To facilitate this, the Commission could clarify that such integrated reforms may fall with the scope of a final rule as “consistent with or superior to” the requirements set forth by the Commission. Recognizing the ability of flexibility commitments to streamline interconnection study requirements and then accounting for those (or related) flexibility commitments in resource adequacy planning appropriately provides speed-to-market for both associated load and supply while keeping rates just and reasonable for existing customers.

Second, the Commission should allow VPPs that meet applicable technical requirements to qualify an associated large load for expedited interconnection service pathways that depend on transmission system impacts. Various parties have proposed different mechanisms to implement the ANOPR’s expedited interconnection and integrated load and generation facility study principles based on the potential for curtailment and dispatch to lead to reduced transmission system impacts. These include technical eligibility requirements that large load customers or hybrid facilities would need to meet in order to be eligible for expedited and integrated interconnection processes. Pew does not take a position on the relative merits of these specific technical proposals. Regardless of which particular suggestion or combination of suggestions the Commission ultimately pursues to enable flexibility, or whether it develops its own approach based on the record, the Commission should adopt resource-neutral definitions that allow VPPs to satisfy technical requirements where they are able to provide the services that the technical requirements are designed to ensure.

³⁹ See Governor Josh Shapiro and The Commonwealth of Pennsylvania, Complaint, Docket No. EL25-46-000, at 17-22 (filed Dec. 30, 2024).

For example, Eolian proposes adoption of serial streamlined study of load- and electrically-close hybrid facilities without carrying out generation and load deliverability analysis where the facility (1) nominates operating limits for aggregate net injections and withdrawals of power, (2) installs necessary control equipment to ensure that the facility is capable of following the dispatch controls of the transmission system operator and balancing authority and respecting operating limits at all times, and (3) is capable at any time of complete self-supply or otherwise curtailing the load such that the facility can take transmission service on a non-firm basis.⁴⁰ VPPs could be compatible with this type of proposal to the extent that contracts or coordinated operation with a jointly-owned portfolio of electrically-close DERs enable the facility to meet the specified operating limits. Where a large load customer's VPP portfolio allows the combined resources to meet those technical requirements, that configuration should enable the same streamlined study. Similarly, Critical Loop urges the Commission to establish a "Large-Load Flexibility Commitment" (LLFC) construct under which transmission providers would be required "to model LLFC parameters explicitly in interconnection studies" for a load that "accepts specified curtailment and/or relies on on-site or adjacent supply under defined conditions"⁴¹ Aggregated DERs, again, should be eligible as "adjacent supply" in this example, because they are as able to reduce the new large load's transmission system impacts as utility-scale resources.

Like utility-scale resources paired with large loads, VPPs could receive credit based on the degree of flexibility they enable. For example, "CAISO recommends that any final rule should establish distinct 'lanes' based on the level of dispatch tools the transmission provider

⁴⁰ Eolian, L.P. Comments at 37, 39.

⁴¹ Critical Loop Comments at 2.

will have—all, some, or none.”⁴² Because “the specific grid topology will . . . be a key factor,”⁴³ eligibility can flow from the particular system location of DERs that form a VPP and from their dispatch capabilities. Importantly, studying VPPs together with large loads through a single integrated process would not mean combining the VPP with the large load and only studying the net service requirements (a simplification that could inappropriately mask reliability risks). Rather, as with generation components of a hybrid facility, the streamlined study process could simply study the VPP and load components together, so that the combined effect of the components on the system become evident while recognizing that they are separate components of a broader configuration.⁴⁴

A key component of enabling these frameworks is to allow large load customers to contract with third parties in order to achieve curtailability or dispatchability. As Base Power recommends, curtailability and dispatchability could “be implemented as defined, contractible commodities in order to be sufficiently integrated into both operations and system planning.”⁴⁵ Base Power proposes that “[t]o be contractible, particularly for dispatchable load, the conditions for operation must be clearly delimited and well-defined, with established parameters for availability and performance requirements (e.g., ramp rate and target power error tolerances).”⁴⁶ Clear definition should include specifying when and how VPPs can meet these requirements. Base Power points to ERCOT’s definition of Controllable Load Resources, which provides one example of a set of resource-neutral performance metrics that could be adopted.⁴⁷

⁴² CAISO Comments at 12.

⁴³ *Id.*

⁴⁴ *See, e.g.,* Eolian, L.P. Comments, Attachment A (setting forth a study methodology that would assess impacts of generation and load components of a hybrid facility as part of a single integrated process)

⁴⁵ Base Power Comments at 3.

⁴⁶ *Id.*

⁴⁷ *Id.* at 4-5.

D. Allowing VPPs to be eligible for hybrid programs would not and should not intrude on states' abilities to incentivize VPPs, including those that simultaneously benefit a FERC-jurisdictional large load.

States have justifiably expressed support for VPPs and wish to retain “the ability to incentivize large loads to invest in local communities, including through programs such as . . . funding virtual power plants, non-transmission alternatives, or demand-side management programs.”⁴⁸ While Pew does not take a position on the question of whether the Commission should exert jurisdiction over large load interconnection, we emphasize that to the extent the Commission does so, including VPPs as eligible to contribute to expedited interconnection through Commission-jurisdictional tariffs would not infringe on the ability of states to require or encourage investment in VPPs as part of any processes or services that remain state jurisdictional, including, for example, large load retail service tariffs. Similarly, rendering VPPs eligible to contribute to large load and hybrid facility interconnection processes would not override the terms of retail VPP programs. Rather, such programs would remain valid as adopted, and VPPs would be eligible to support FERC-jurisdictional interconnection service while simultaneously participating in such a program only to the extent the commitments provided pursuant to the relevant state program were consistent with the obligations of the VPP pursuant to the FERC-jurisdictional interconnection agreement of the applicable large load customer or hybrid facility. Importantly, because FERC’s rule in this context would be to establish an interconnection process yielding an interconnection agreement and not to *compensate* resources, the risk of double counting addressed in FERC Order No. 2222 is not presented in the same manner in this context. However, to the extent that any FERC regime involves assessing the net

⁴⁸ National Association of Regulatory Utility Commissioners Comments, “Resolution Urging the Federal Energy Regulatory Commission to Preserve and Affirm State Retail Regulatory Jurisdiction in its Large Load Interconnection Proceeding.”

impacts of the load plus the VPP, FERC should require transmission providers to ensure that the VPP's load reduction potential has not already been factored into the load forecast through a separate process. In other words, the Commission should ensure that modeling and other aspects of the interconnection process do not inappropriately double count the impact of the VPP.

II. CONCLUSION

VPPs are critically positioned to allow the timely interconnection of large loads while protecting existing customers from adverse impacts and allowing for additional benefits. They are especially well positioned to bring capacity online quickly given their rapid deployment speed and ability to enable load to avoid the need for network upgrades. Implementing large load and hybrid facility interconnection rules in a resource-neutral manner that allows for VPP eligibility is consistent with Commission precedent and critical to bringing every possible megawatt able to serve growing electricity demand online as fast as possible. If the Commission asserts jurisdiction in this rulemaking proceeding, Pew urges the Commission to draft its rules in such a way that unlocks the multiple benefits that VPPs can provide both to existing transmission customers and large load customers seeking to interconnect while keeping electricity rates affordable and maintaining grid reliability.

Respectfully submitted,

A handwritten signature in cursive script, reading "Maureen Quinlan". The signature is written in dark ink on a light blue background.

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